
Workshop on Usage Control Policies in Dataspaces

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Agenda



- **Intro:** What are Dataspaces
- **Dataspace Connectors:** A review of existing Implementations
- **Use Case:** “Data-driven Tourism for Sustainability”
- **Demo:** nexyo data Hub and Ecosystem
- **Usage Control Policies:** A Review of Policy Classes

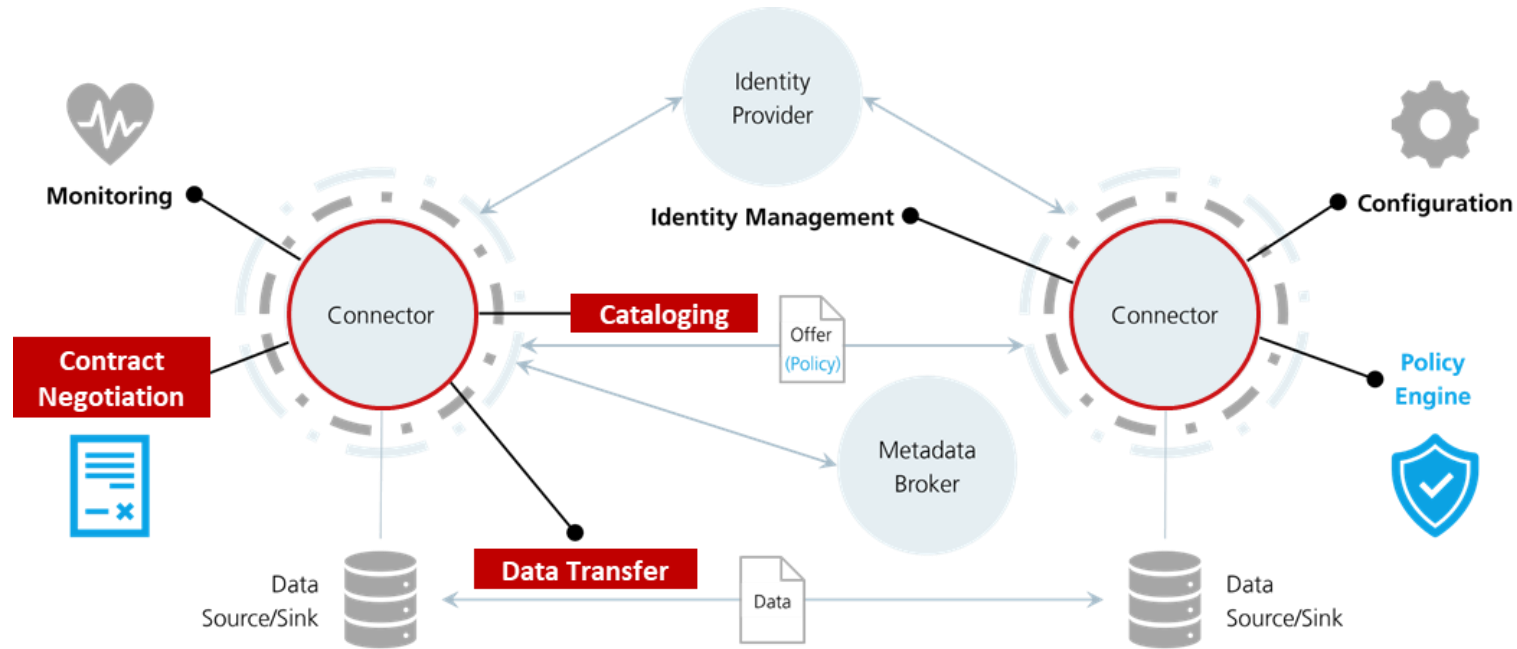
European Dataspaces – Gaia-X

- European project working on the federation of data infrastructure and service providers
- Defines set of *technical and organisational standards* for storage and exchange
- „*Dataspaces*“ as relationship between trusted partners
- Existing reference architecture model for participation by the *International Data Spaces Association (IDSA)*

Data Spaces

- Data spaces allow organizations to securely share data with other participants
- Data are not stored centrally, only transferred (through semantic interoperability) as necessary
- Data spaces are built on
 - **Identity**: Each participant remains in control of their identity
 - **Trust**: Each participant decides who to trust
 - **Sovereignty**: Each participant decides under what policies their data is shared
 - **Interoperability**: Each participant remains in control of their deployment

Architecture



- IDS Information Model published by IDSA
 - RDFS/OWL ontology:

International Data Spaces Information Model

Release 2022-09-16

This version:

<https://w3id.org/idsa/core-420>

Latest version:

<https://w3id.org/idsa/core>

Previous version:

<https://w3id.org/idsa/core-410>

Revision:

4.2.0

A Survey of Dataspace Connector Implementations

IDS Dataspace Connector

soivity, initially Fraunhofer ISST

- IDS Usage Control Language based on ODRL
- Defines 21 policy classes for Usage Control enforcement – 9 are implemented

Eclipse Dataspace Connector (EDC)

Consortium: Microsoft, BMW, Fraunhofer, SAP, etc.

- ODRL-based policy definition, however, currently in Java code

TRUsted Engineering Connector

Engineering Ingegneria Informatica S.p.A

- Platoon/MyData app (using the IDS language)
- Modification of the IDS Connector

Trusted Connector

Fraunhofer AISEC

- LUCON policy language
- Consist of flow rules and a service descriptions

Development Process / Maturity

Connector	Created	Stars	Commits	Language	License	Usage control
DSC	2020-10-07	16	2600	Java	Apache License 2.0	IDS Usage Control Language (based on ODRL)
EDC	2021-07-26	134	1311	Java	Apache License 2.0	Policies defined in connector code.
TRUE	2020-10-30	13	77	Java	AGPLv3	Platoon/MyData app (using the IDS language)
Trusted	2017-09-05	3	2083	Kotlin	Apache License 2.0	LUCON policy language

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Eclipse Data Space Connector

- Eclipse Dataspace Connector (EDC) provides a framework for dataspace connectors
- Contains modules for data discovery, data exchange, policy enforcement, monitoring and auditing
- Implements decentralized identifiers (DIDs) and uses the DID:Web method

DIDs for Identity Management

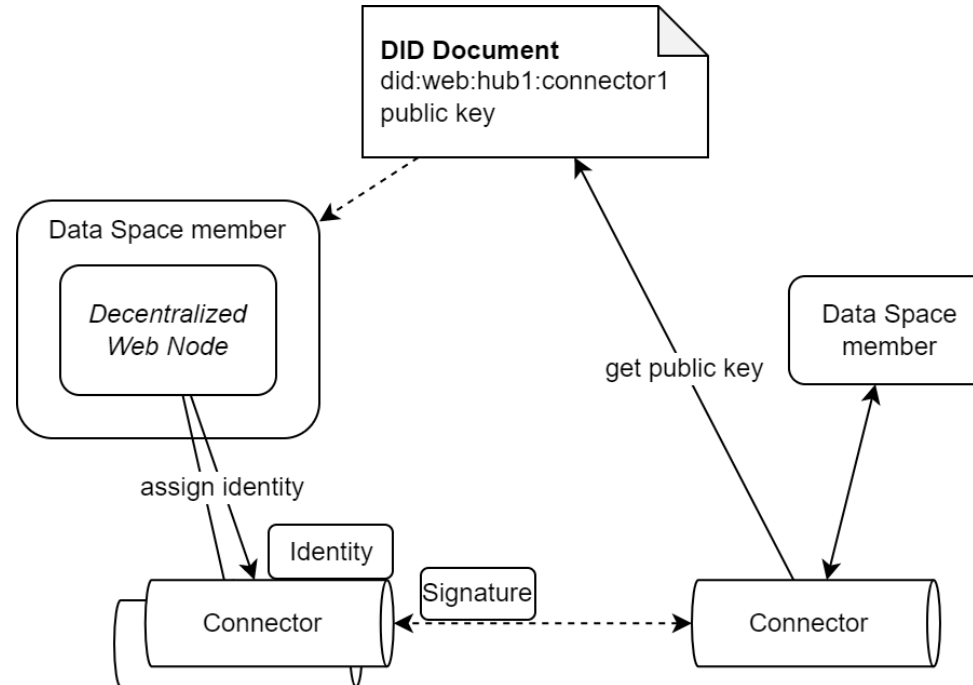
- Decentralized identifier (DID)
 - W3C Proposed Recommendation (w3.org/TR/did-core/)

E.g., 

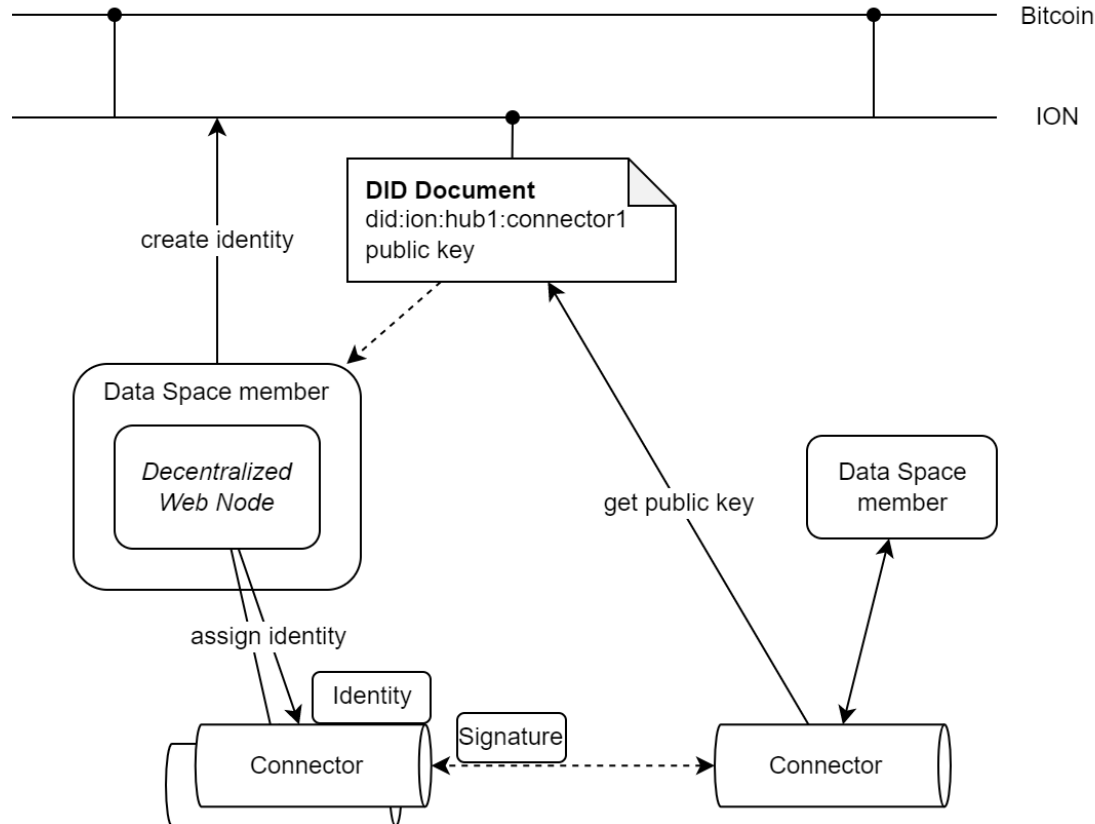
- Resolves to a DID document
 - simple JSON file that contains information

```
{
  "@context": [
    "https://www.w3.org/ns/did/v1",
    "https://w3id.org/security/suites/ed25519-2020/v1"
  ],
  "id": "did:example:123456789abcdefghi",
  "authentication": [{
    // used to authenticate as did:...fghi
    "id": "did:example:123456789abcdefghi#keys-1",
    "type": "Ed25519VerificationKey2020",
    "controller": "did:example:123456789abcdefghi",
    "publicKeyMultibase": "zH3C2AVvLMv6gmMnam3uVAjZpfkcJCwDwnZn6z3wXmqPV"
  }]
}
```

DIDs in a Data Space Architecture



DIDs in a Data Space Architecture (cont'd)



Usage Control Policies in Dataspaces

- Survey existing projects/literature on concrete examples and implementations of usage control policies
- Criteria:
 - Filter policies for Dataspace use cases
 - Potentially enforceable
 - Defined between the interacting parties:
 - Provider, consumer, third-party / observer

General Policies	
Allow access (e.g. for specific user or connector)	[1,2]
Location / Regional restriction	[1,2,3]
Time restriction (e.g. start, end, duration)	[1]
Access count	[1,2]
Rate limit (e.g., downloads within hour)	[4]
Number of concurrent active connections	[4]
Fixed amount of data (e.g. in streaming)	[2]
Processing power	[2,9]
Usage of bandwidth	[2,9]
Billing / Credit points (perpetual vs. rental)	[1,2]

Data Usage	
Deletion	[1,5]
Purpose / Application	[1]

Attribute based / Obligation fulfilled	
Proofable attributes, e.g. membership credentials, position in company, security clearance/certification	[1,6]
Non-proofable, e.g. legal jurisdiction	[4,6]

Privacy Policies	
Encryption: at rest / at transfer	[1,3]
Aggregation	[7]
Anonymization	[7,10]

Traceability	
Logging	[1,5]
Notification	[1]
Delegation/distribution of permissions to third-party	[1,2]

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